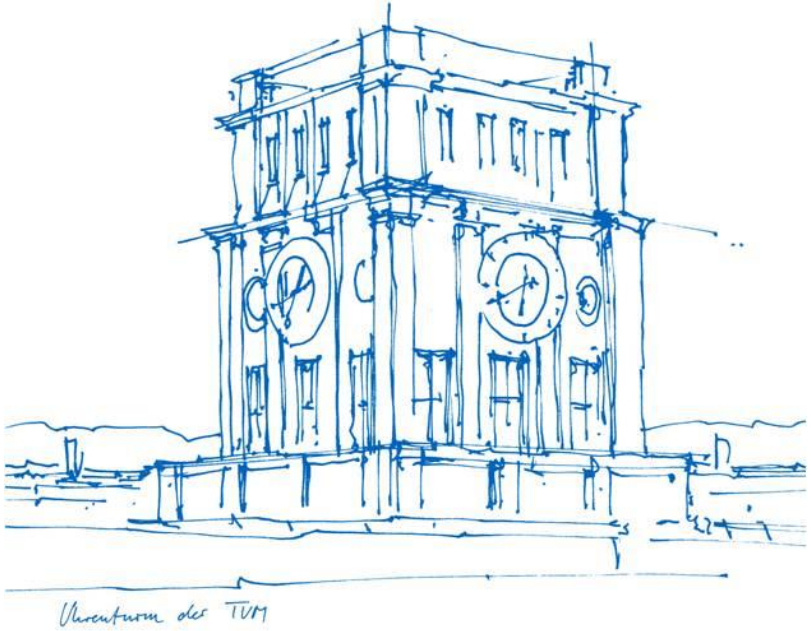
Lehrstuhl für Rechnerarchitektur & Parallele Systeme

# Grundlagenpraktikum: Rechnerarchitektur

## Sobel-Filter (A201)

Salim Daoud - Tobias Langer - Luca Tänzler




Technische Universität München
TUM School of Computation, Information and Technology
Wintersemester 24/25

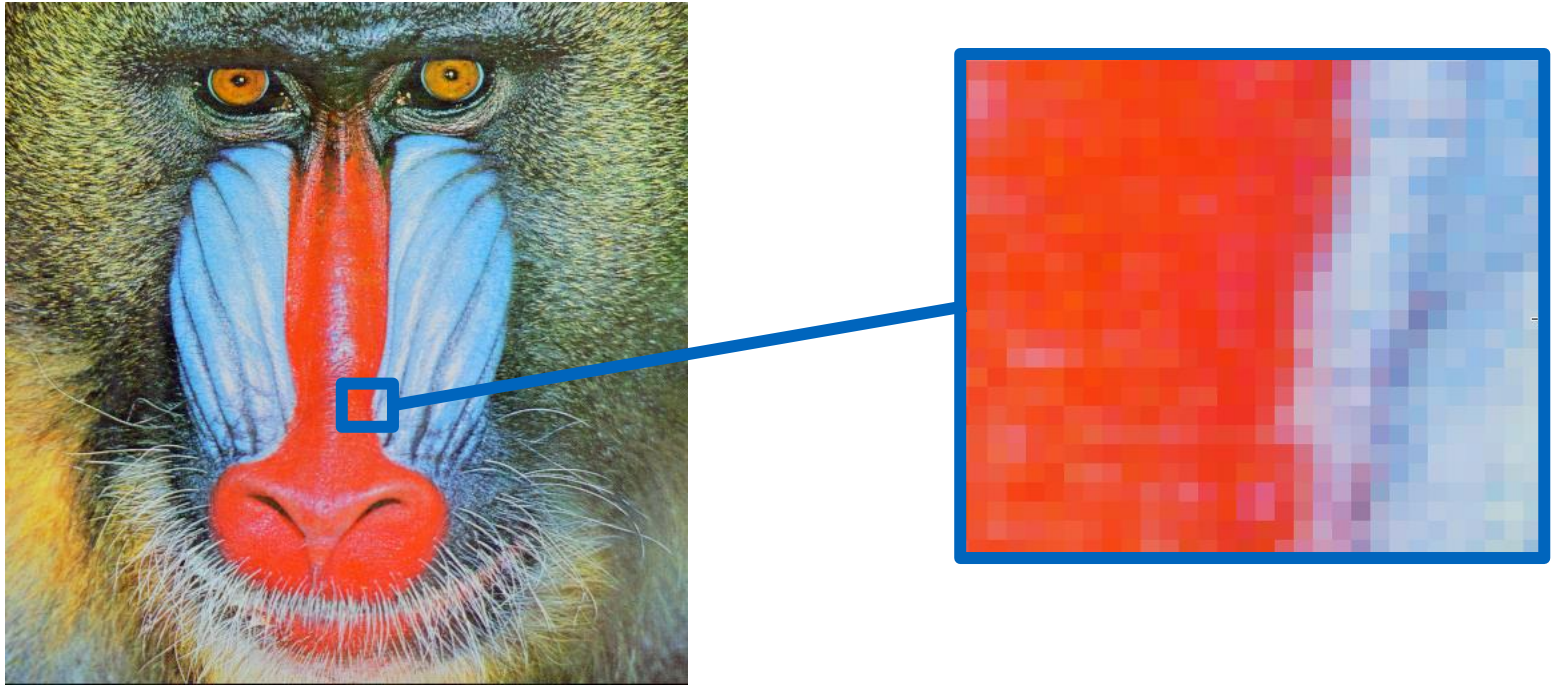
# Content

1. Problem Statement
2. Solutions and Optimizations
3. Correctness
4. Performance Analysis
5. Summary and Outlook

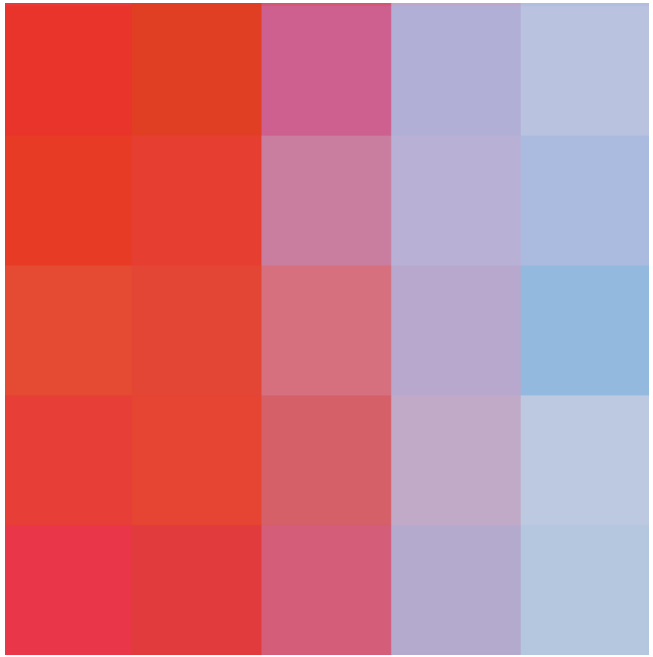
# Implement edge detection using Sobel-Filter in C

- Grayscale Conversion
- Edge Detection with Sobel-Filter
- Optimize Computation
- Performance Evaluation

# Detecting edges

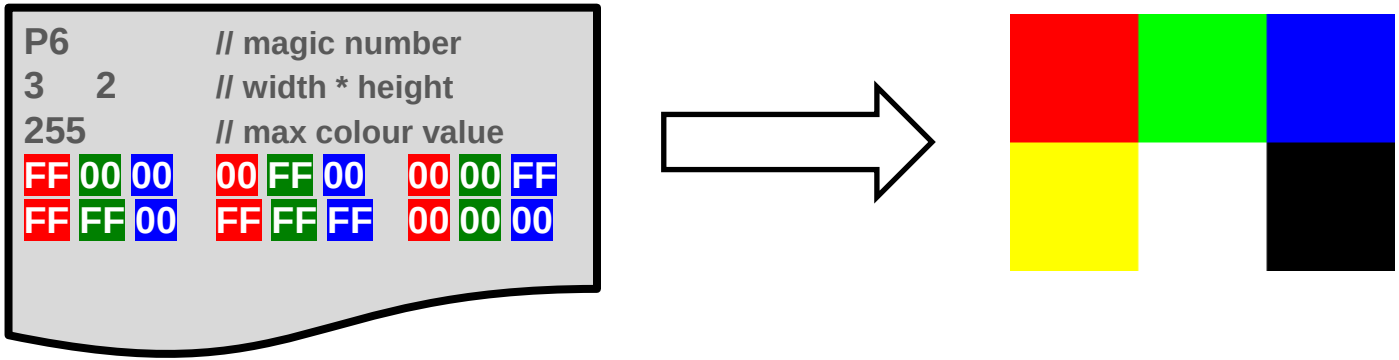


# Detecting edges



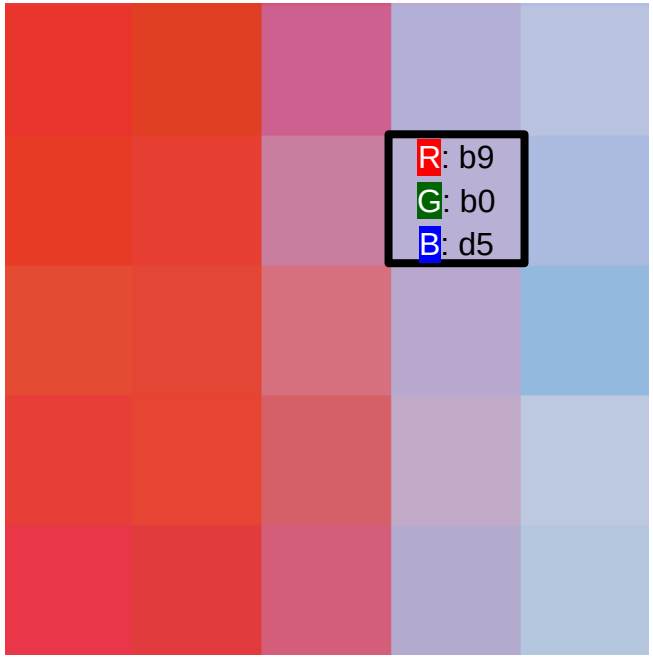
# Image Format: PPM & PGM

PPM is used for storing color images in simple, binary format:




PGM is used for grayscale images, storing only one intensity value per pixel instead of three.

# Grayscaling




Basic: mean value

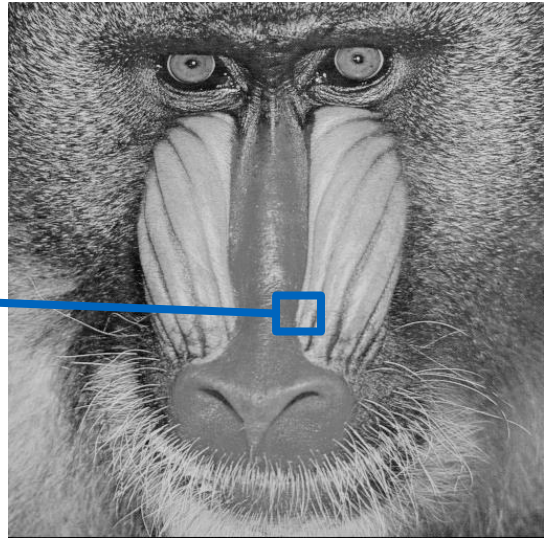
$$\boxed{G = \frac{R+G+B}{3}}$$

Adapted to the human eye: weighted values:

$$\boxed{G = R * 0.299 + G * 0.587 + B * 0.114}$$

# Grayscaling

| | | | | |
|---|---|---|---|---|
| 104 | 108 | 134 | 180 | 194 |
| 108 | 111 | 152 | 182 | 186 |
| 118 | 114 | 144 | 177 | 178 |
| 111 | 115 | 132 | 180 | 199 |
| 111 | 109 | 131 | 176 | 196 |



# Sobel Filter - Formal Description

| | | | | |
|---|---|---|---|---|
| 104 | 108 | 134 | 180 | 194 |
| 108 | 111 | 152 | 182 | 186 |
| 118 | 114 | 144 | 177 | 178 |
| 111 | 115 | 132 | 180 | 199 |
| 111 | 109 | 131 | 176 | 196 |

2 Sobel Kernels:

$$M^v = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{pmatrix} \qquad M^h = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix}$$

Convolution:

$$Q^d_{(x,y)} = \sum_{i=-1}^{1} \sum_{j=-1}^{1} M^d_{(1+i,1+j)} \cdot Q_{(x+i,y+j)} \qquad d \in \{v, h\}$$

Combining $Q^v$ and $Q^h$

$$Q'_{(x,y)} = clamp_0^{255}\left(\sqrt{\left(Q^v_{(x,y)}\right)^2 + \left(Q^h_{(x,y)}\right)^2}\right)$$

# Naive Implementation

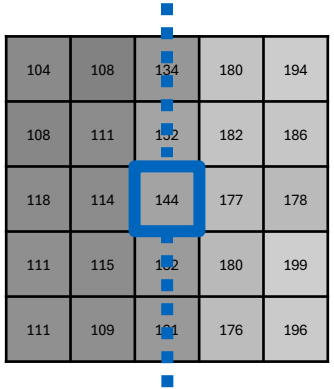
| | | | | |
|---|---|---|---|---|
| 104 | 108 | 134 | 180 | 194 |
| 108 | 111 | 132 | 182 | 186 |
| 118 | 114 | 144 | 177 | 178 |
| 111 | 115 | 132 | 180 | 199 |
| 111 | 109 | 131 | 176 | 196 |

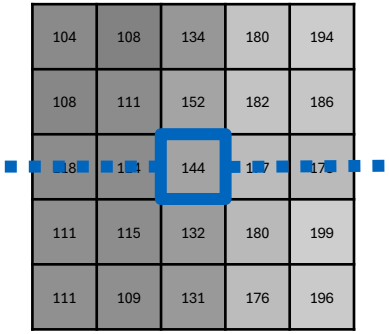
Applying Vertical Kernel:

```
int kern_vertical[3][3] = {    {1, 0, -1},
                               {2, 0, -2},
                               {1, 0, -1} };
```

111 + 2 * 114 + 115
+ 0 * 152 + 0* 144 + 0* 132 = - 262
- (182 + 2 * 177 + 180)

# Naive Implementation



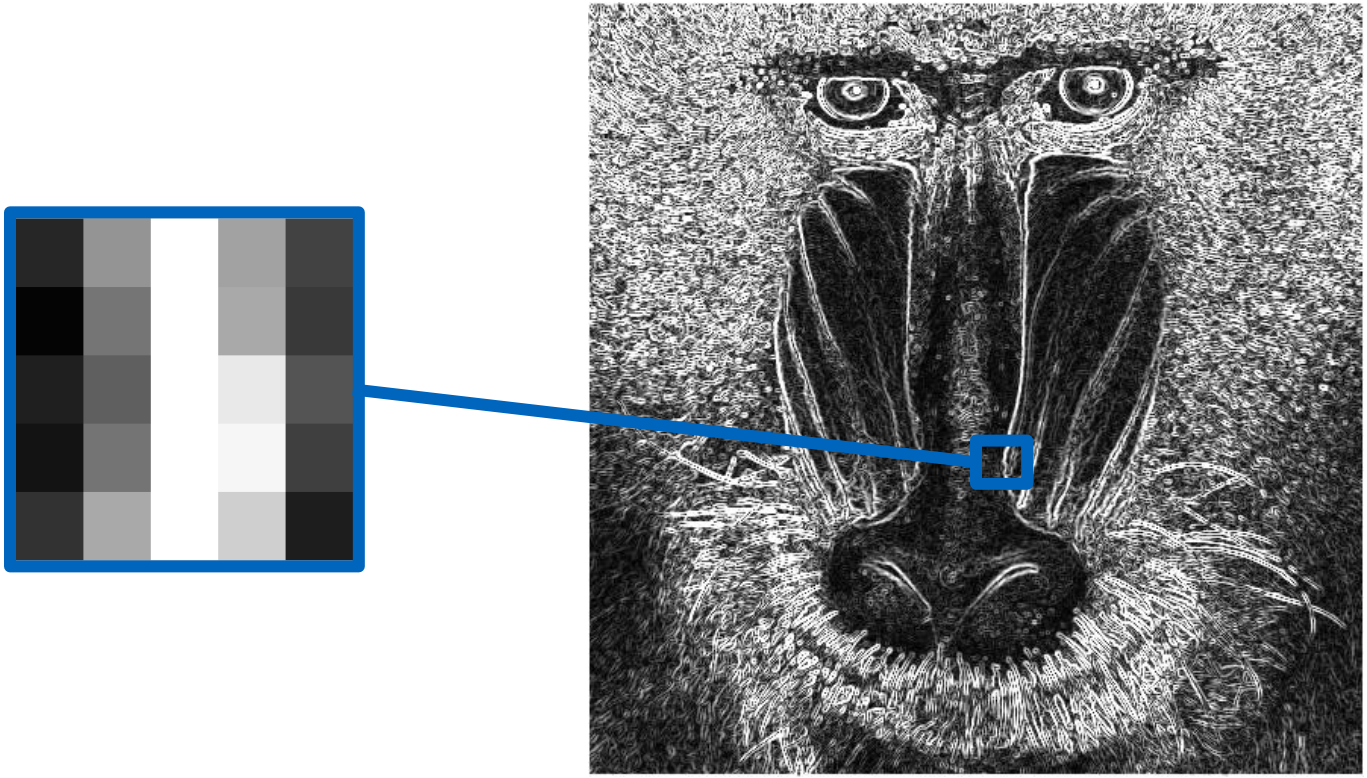




```
int magnitude = (int) sqrt(sum_vertical * sum_vertical + sum_horizontal * sum_horizontal);
```

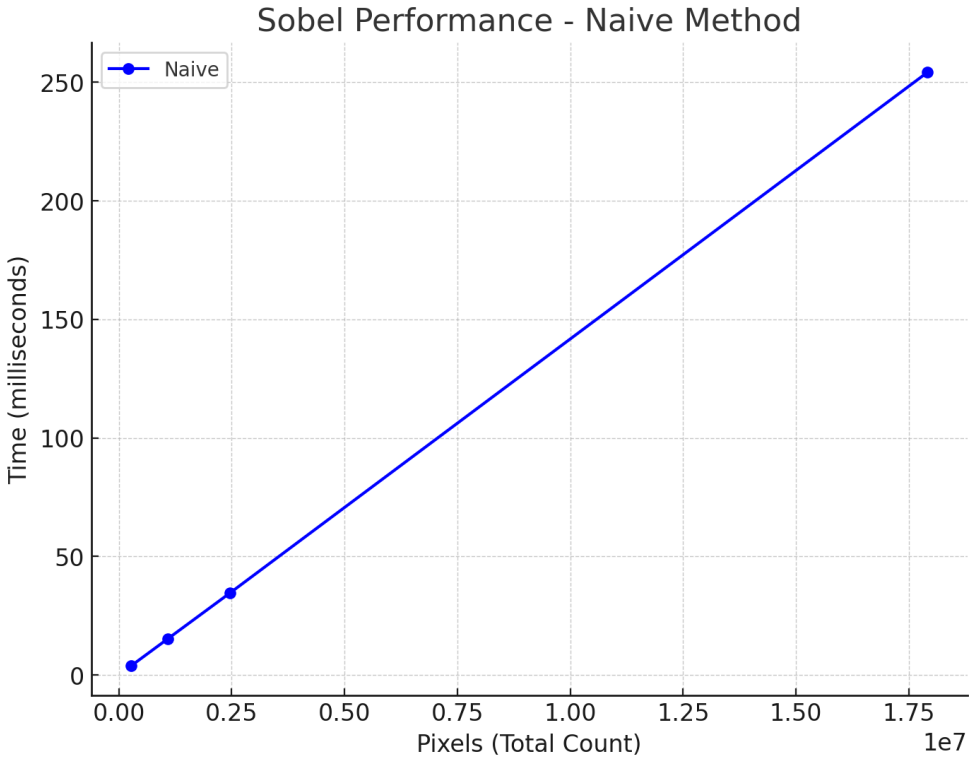
# Result

| | | | | |
|---|---|---|---|---|
| 38 | 148 | 255 | 162 | 66 |
| 4 | 117 | 255 | 169 | 57 |
| 31 | 95 | 255 | 233 | 84 |
| 19 | 116 | 255 | 246 | 63 |
| 50 | 169 | 255 | 207 | 29 |

# Result



# Sobel naive performance




AMD Ryzen 5 4680U, 2.20 GHz, 8 GB RAM, compiled with -O2

# Sobel Kernel Unroll

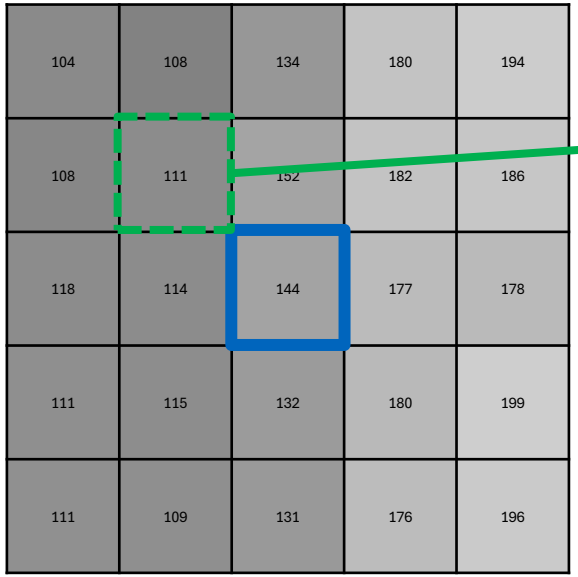
$$M^v = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{pmatrix}$$

$$M^h = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix}$$

Iteration over 2 Kernels to fetch scale factors is inefficient, but:

- fixed, small size
- both Kernels address same same pixels
- multiplication with 0

# Sobel Kernel Unroll



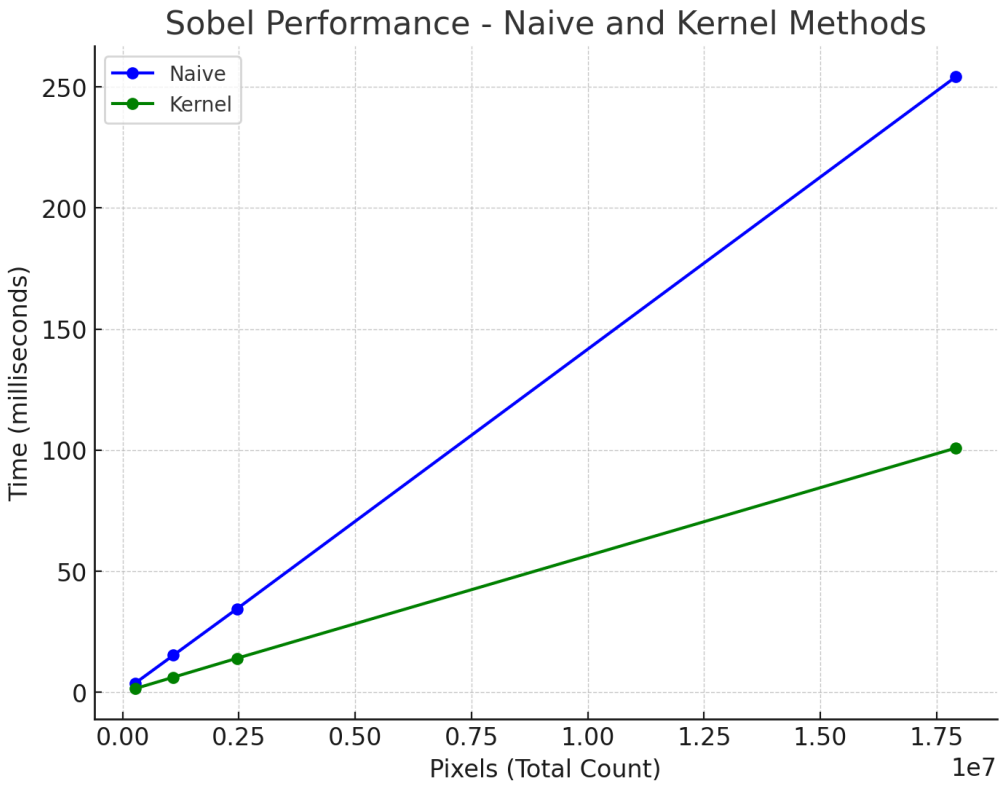

buffer and reuse neighbouring pixels

```
uint8_t top_left =  image_row_prev[gray_x - 1];
```

hardcode formula per Kernel

```
gradient_vert = top_left + left + left + bot_left
                - top_right - right - right - bot_right;
```

# Sobel Kernel Unroll performance




60% faster than Naive

AMD Ryzen 5 4680U, 2.20 GHz, 8 GB RAM, compiled with -O2

# Sobel Separate Convolution

right neigbour:

$$[\ldots] - \begin{bmatrix} 1 * 152 \\ 2 * 144 \\ 1 * 132 \end{bmatrix}$$

| | | | | |
|---|---|---|---|---|
| 104 | 108 | 134 | 180 | 194 |
| 108 | 111 | 152 | 182 | 186 |
| 113 | 114 | 144 | 177 | 178 |
| 111 | 115 | 132 | 180 | 199 |
| 111 | 109 | 131 | 176 | 196 |

left neighbour:

$$[\ldots] + \begin{bmatrix} 1 * 152 \\ 2 * 144 \\ 1 * 132 \end{bmatrix}$$

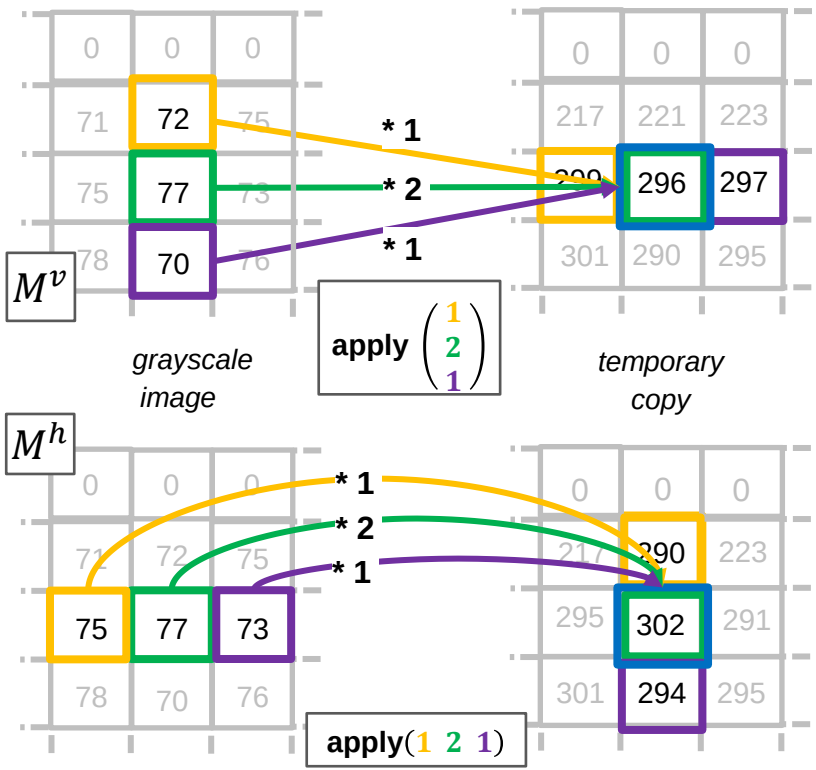
# Sobel Separated Convolution

- The Sobel filter matrix is separable.
- Horizontal and vertical kernel can be displayed as the outer product of two 1-dimensional vectors:

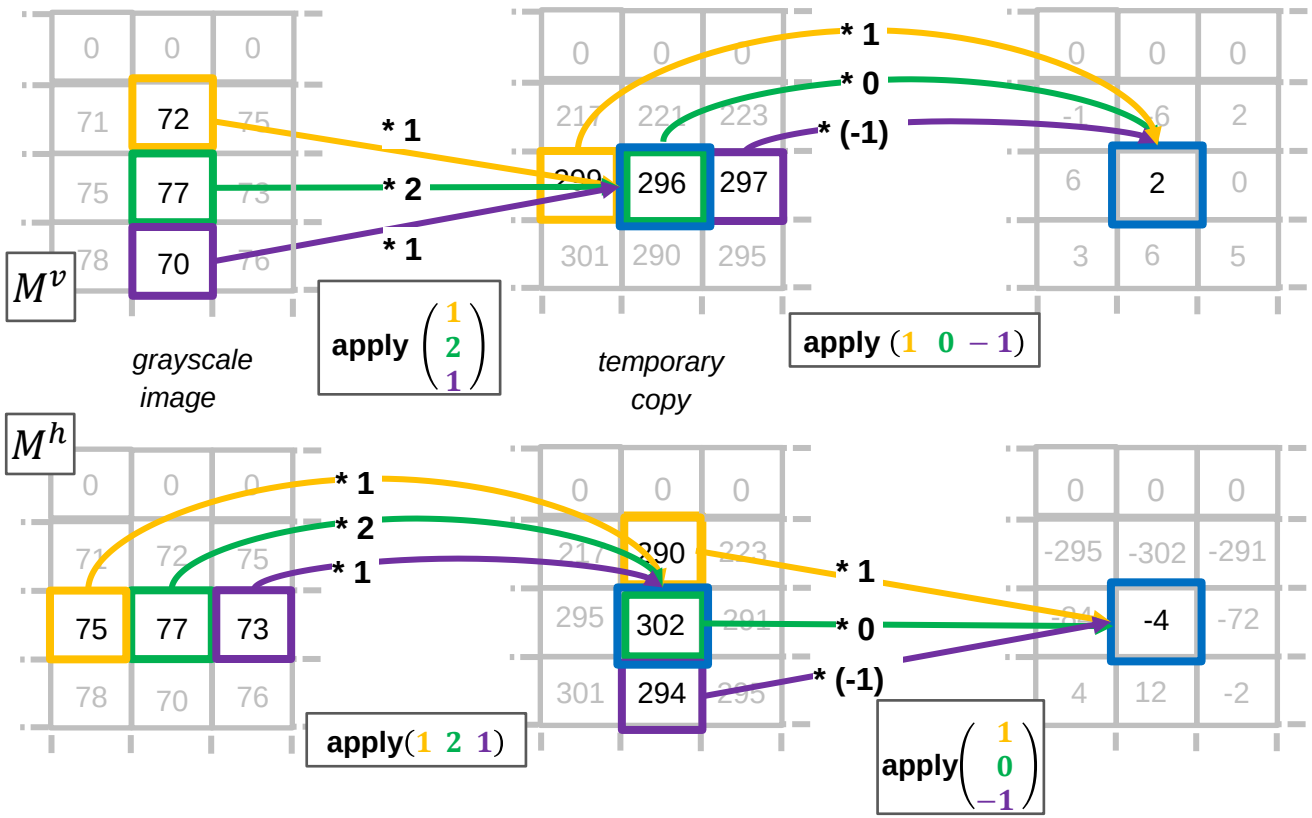
$$M^v = \begin{pmatrix} 1 & 0 & -1 \\ 2 & 0 & -2 \\ 1 & 0 & -1 \end{pmatrix} = \begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix} * \begin{pmatrix} 1 & 0 & -1 \end{pmatrix}$$

$$M^h = \begin{pmatrix} 1 & 2 & 1 \\ 0 & 0 & 0 \\ -1 & -2 & -1 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 1 \end{pmatrix} * \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$$

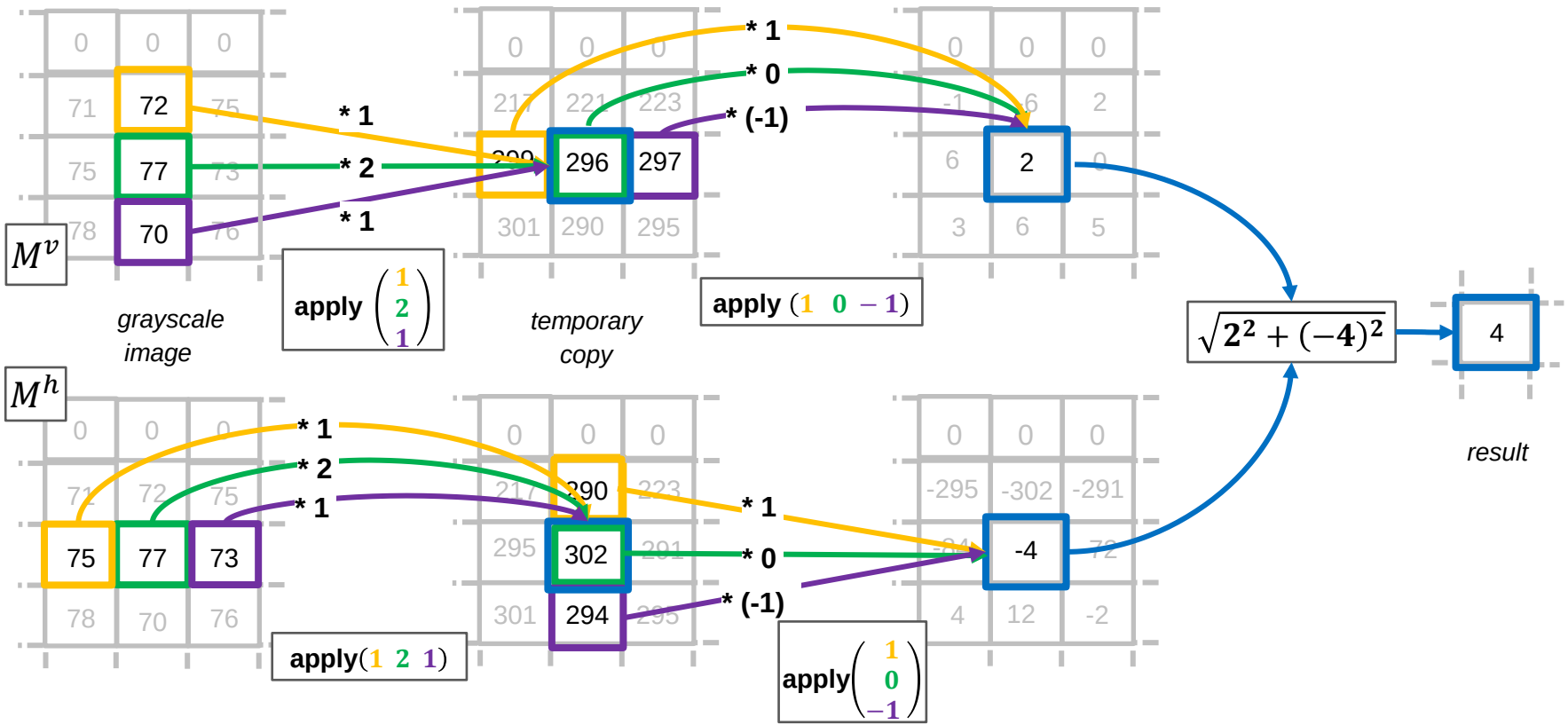
# Sobel Separate Convolution



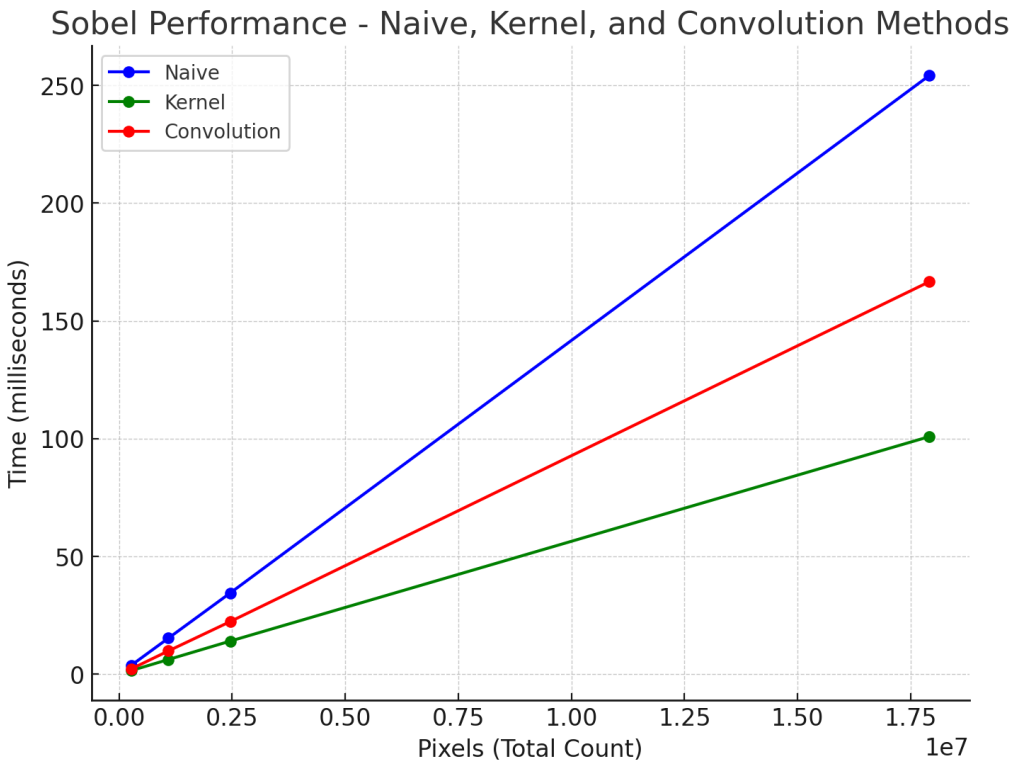
# Sobel Separate Convolution



# Sobel Separate Convolution



# Sobel Seperated Convolution performance

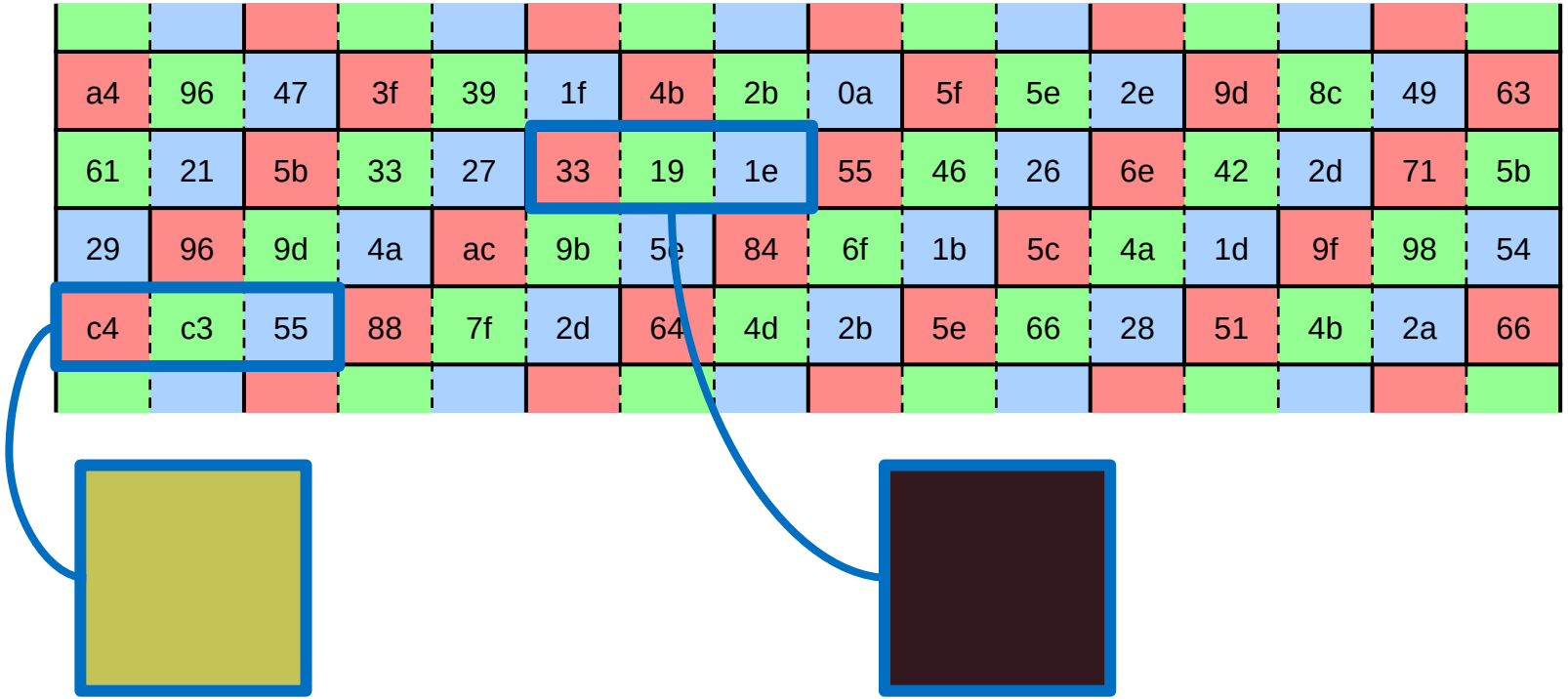



34% faster than Naive

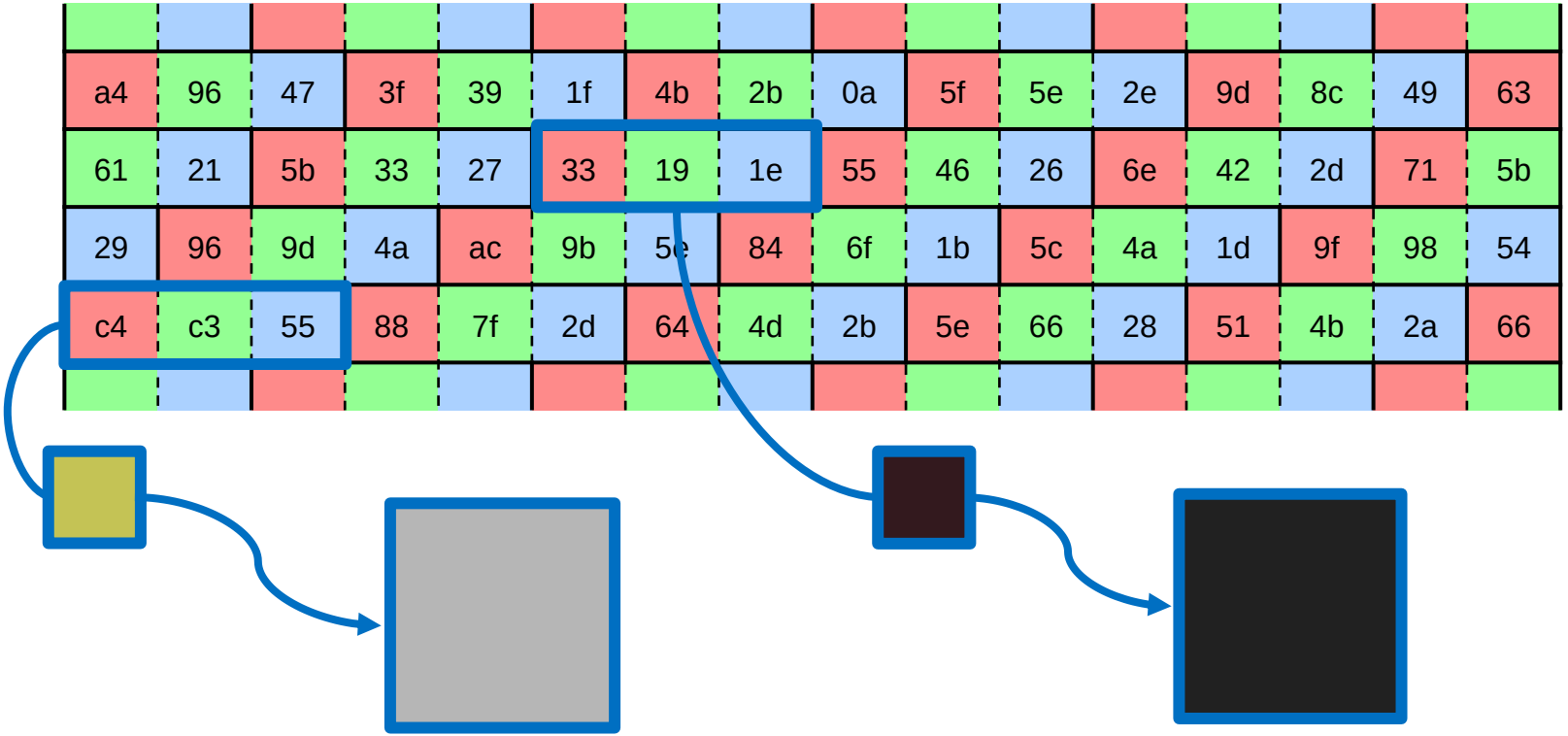
But 39% slower than Kernel unroll

AMD Ryzen 5 4680U, 2.20 GHz, 8 GB RAM, compiled with -O2

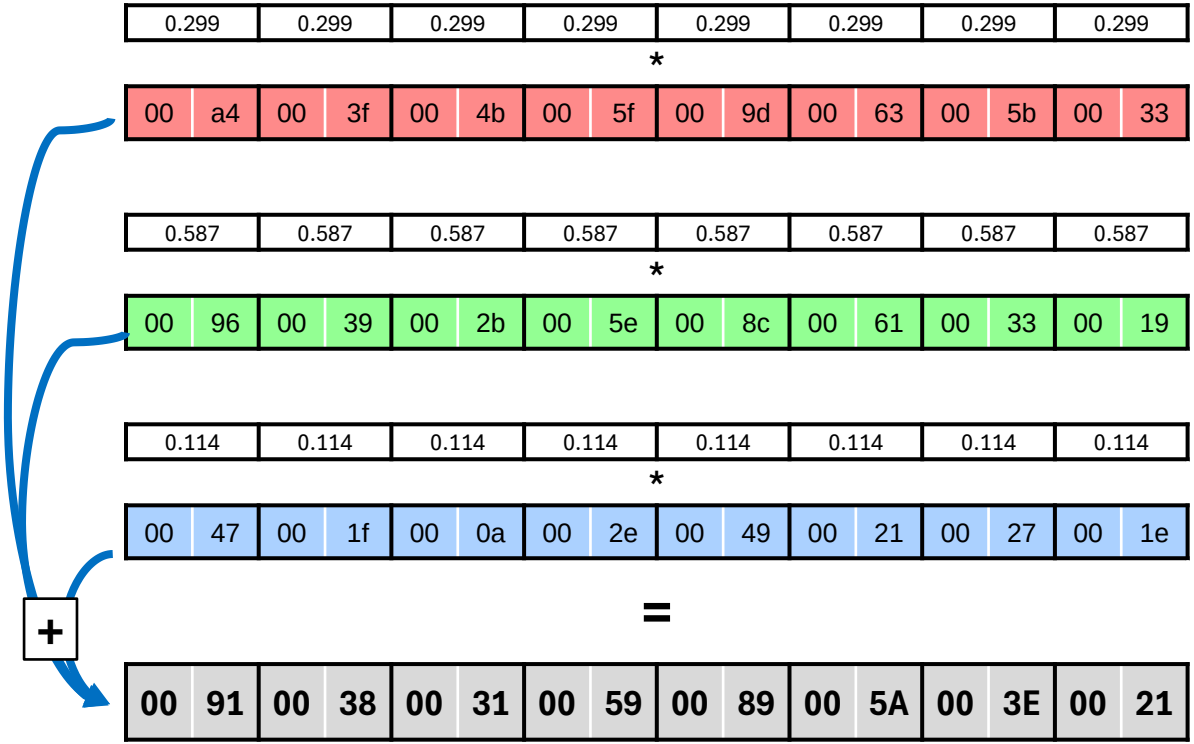
# RGB to Grayscale



# RGB to Grayscale



# RGB to Grayscale SIMD – basic idea




Obstacles:

- Floating point multiplication needs 32 Bit
- RGB values not stored as a sequence

# Grayscaling 8 pixels with SIMD

| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| a4 | 96 | 47 | 3f | 39 | 1f | 4b | 2b | 0a | 5f | 5e | 2e | 9d | 8c | 49 | 63 |
| 61 | 21 | 5b | 33 | 27 | 33 | 19 | 1e | 55 | 46 | 26 | 6e | 42 | 2d | 71 | 5b |
| 29 | 96 | 9d | 4a | ac | 9b | 5e | 84 | 6f | 1b | 5c | 4a | 1d | 9f | 98 | 54 |
| c4 | c3 | 55 | 88 | 7f | 2d | 64 | 4d | 2b | 5e | 66 | 28 | 51 | 4b | 2a | 66 |

*_mm_loadu_si128*(( m128i*)( image_pointer ))

**reg1:**

| | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 5f | 5e | 2e | 9d | 8c | 49 | 63 | 61 | 21 | 5b | 33 | 27 | 33 | 19 | 1e | 55 |
| 0 | | | 1 | | | 2 | | | 3 | | | 4 | | |

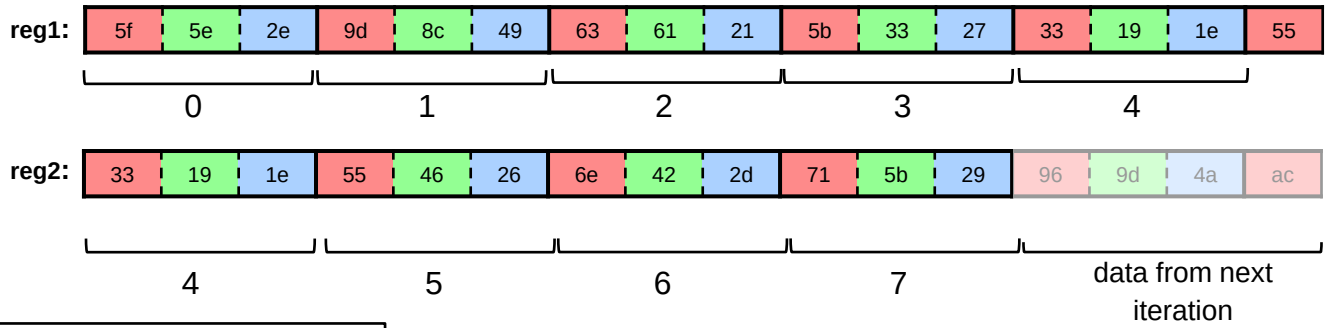
*_mm_loadu_si128*(( m128i*)( image_pointer + 12 ))

**reg2:**

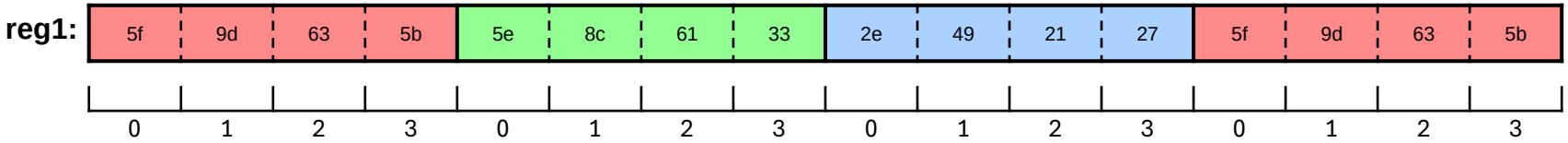
| | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 33 | 19 | 1e | 55 | 46 | 26 | 6e | 42 | 2d | 71 | 5b | 29 | 96 | 9d | 4a | ac |
| 4 | | | 5 | | | 6 | | | 7 | | | data from next iteration | | | |

8 values can be computed using 1 *__m128i* register and fixed point arithmetic.

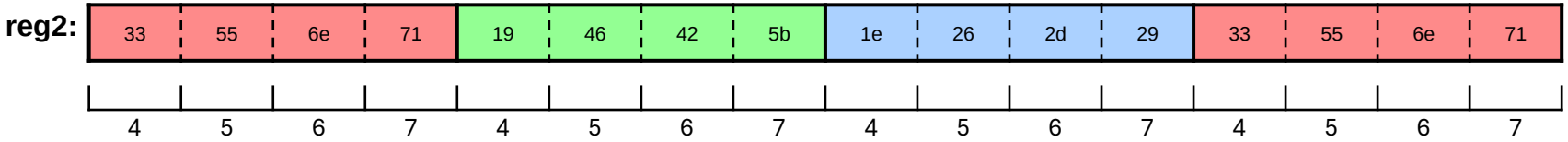
# Grouping the RGB data



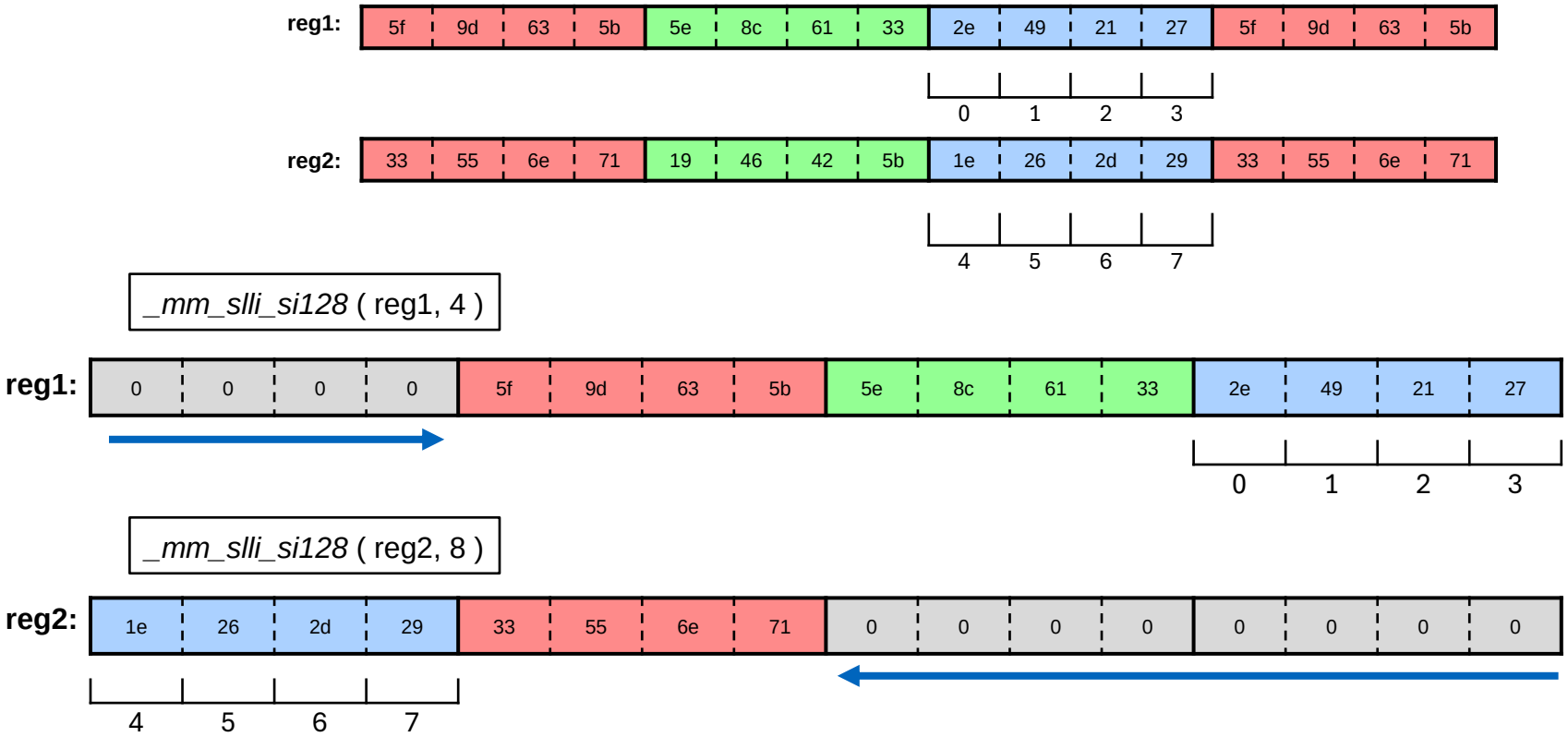

*_mm_shuffle_epi8*( reg1, mask_0123 )



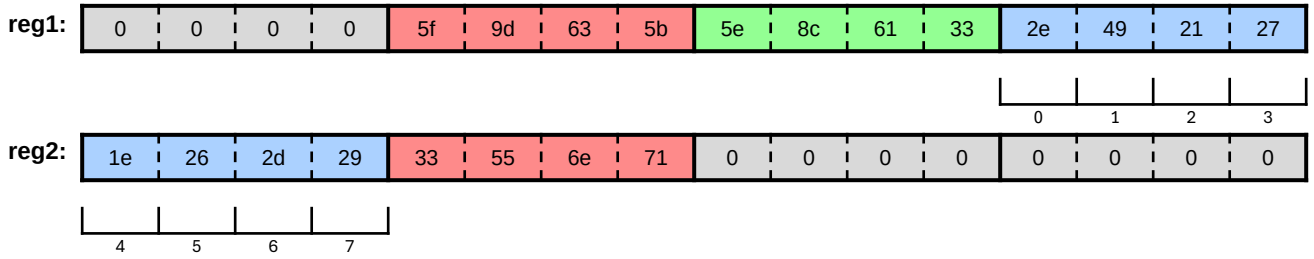

*_mm_shuffle_epi8*( reg2, mask_4567 )



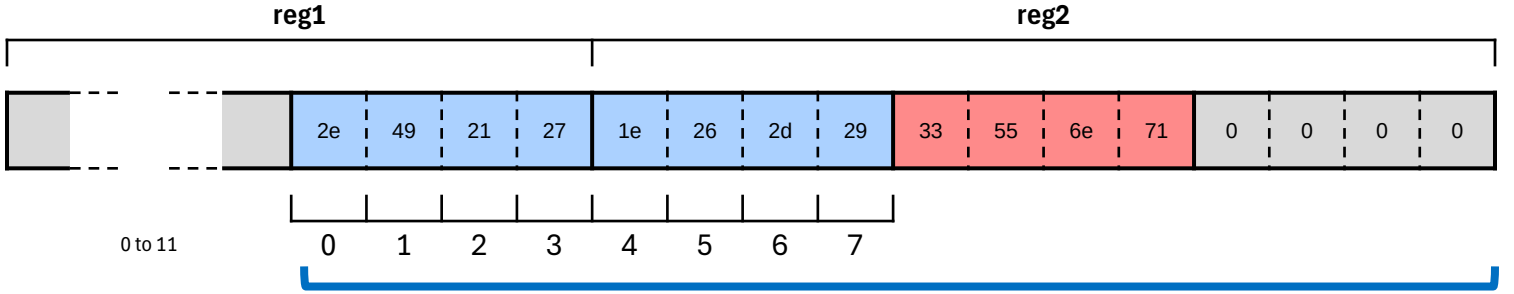
# Isolating single colours



# Extracting single colours

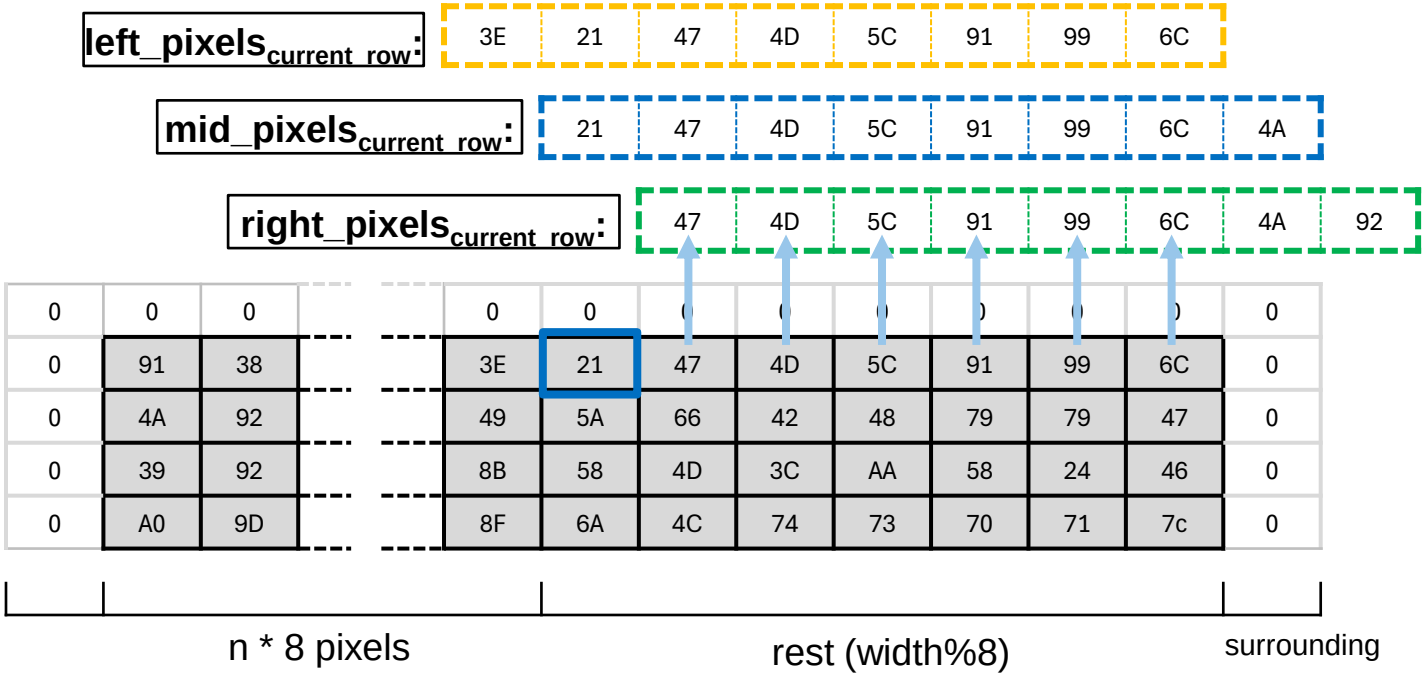



*_mm_alignr_epi8*( reg1, reg2, 12 )

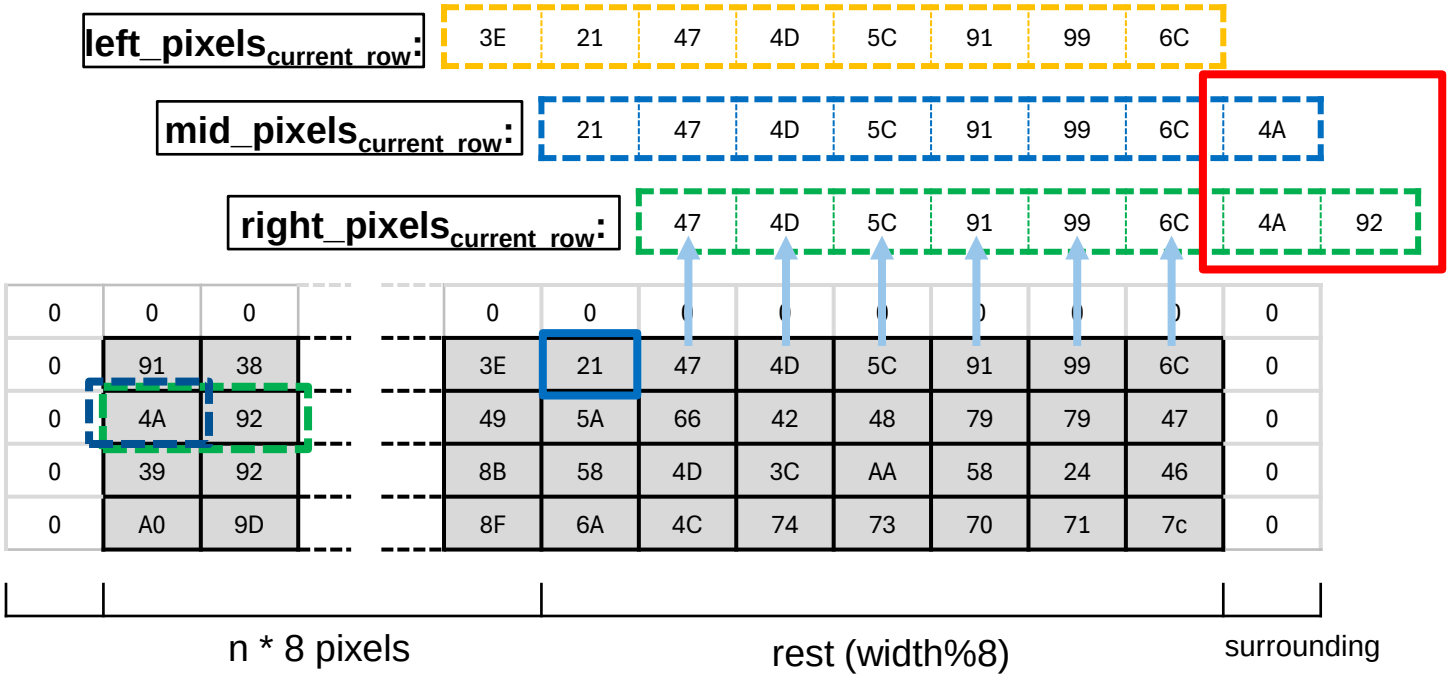



extend to 16 Bit values and apply respective coefficient

# Sobel SIMD


left_pixels current_row:
3E 21 47 4D 5C 91 99 6C
mid_pixels current_row:
21 47 4D 5C 91 99 6C 4A
right_pixels current_row:
47 4D 5C 91 99 6C 4A 92
0 0 0 0 0 0 0 0 0 0 0 0 0
0 91 38 3E 21 47 4D 5C 91 99 6C 0
0 4A 92 49 5A 66 42 48 79 79 47 0
0 39 92 8B 58 4D 3C AA 58 24 46 0
0 A0 9D 8F 6A 4C 74 73 70 71 7c 0
n * 8 pixels
rest (width%8)
surrounding

# Sobel SIMD



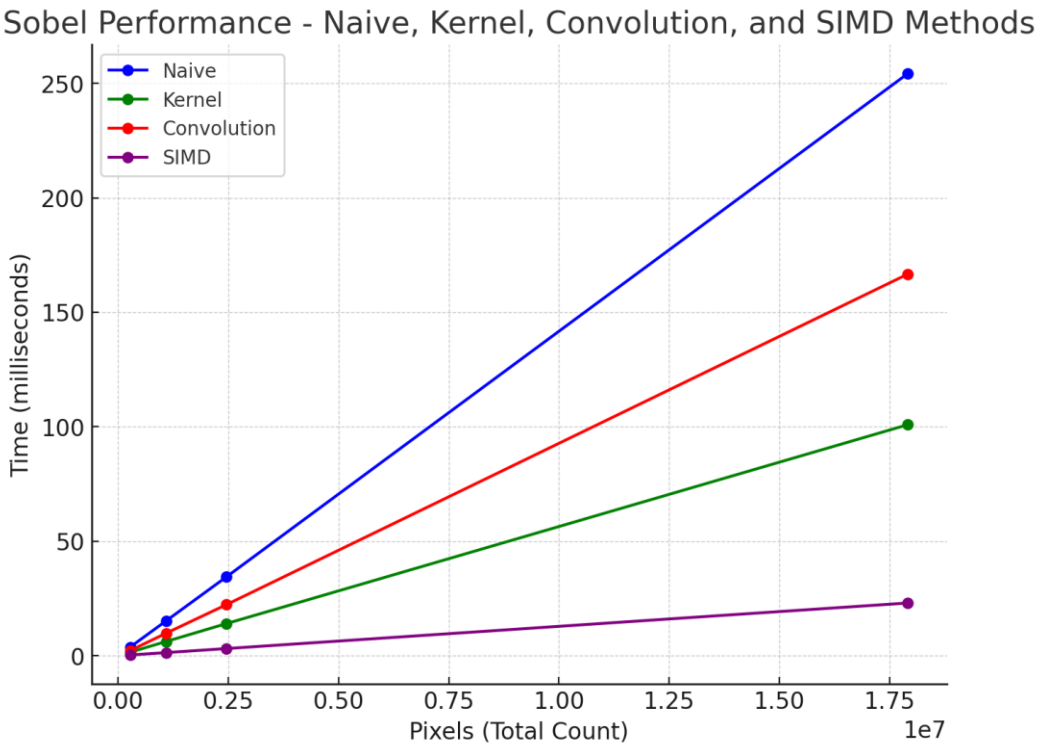

**Problem:**

loading 8 values at once leads to overlapping into the next rows

**Solution:**

Masking the loaded values depending on the number of remaining pixels per row.

# Sobel SIMD performance




90% faster than Naive

77% faster than Kernel unroll

86% faster than Convolution

AMD Ryzen 5 4680U, 2.20 GHz, 8 GB RAM, compiled with -O2

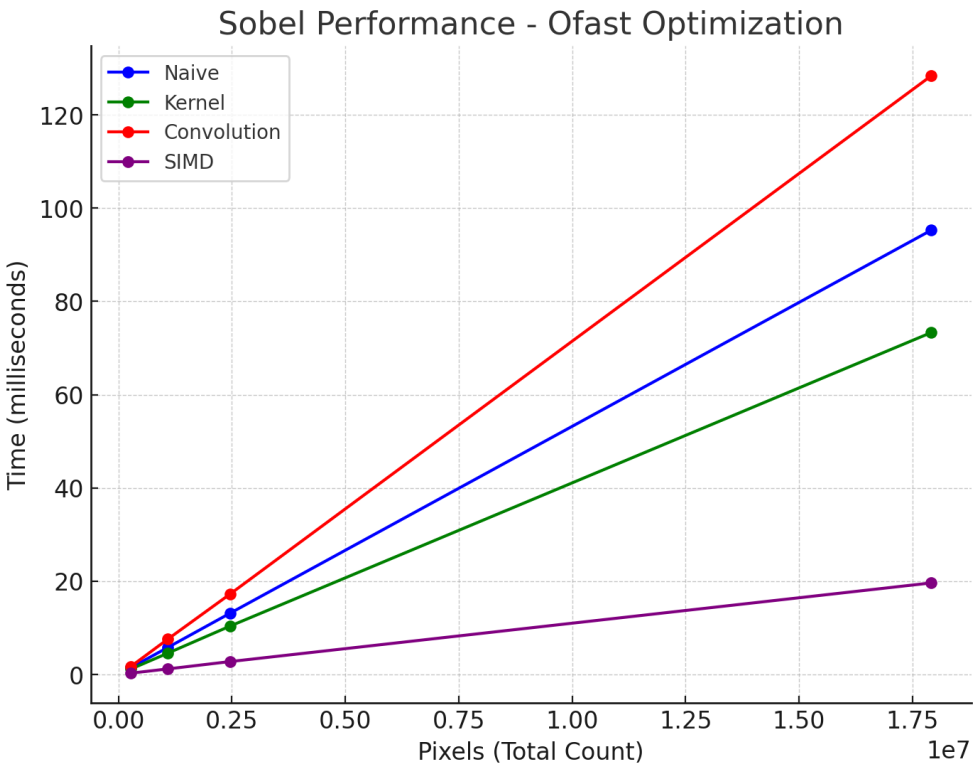
# Unit tests

```
==========================================================================================
SUCCESSFUL:  Grayscale SIMD Basic Test
==========================================================================================
SUCCESSFUL:  Grayscale Basic Test
==========================================================================================
WARNING: The grayscale coefficients are only roughly approximated: Red (0.25), Green (0.5) Blue (0.25).
Given coefficients for Red (0.299), Green (0.587) and Blue (0.114) will not be used.
FAILED:      Grayscale Bitshift Basic Test
             Result does not match expectations, see test_img_to_grayscale_bitshift() in test/grayscale_test.c
             Details:
                          Expected:    Got:
                     [ 0]       203     189
                     [ 1]       165     154
                     [ 2]       117     120
                     [ 3]       134     151
                     [ 4]        97     113
                     [ 5]       164     150
                     [ 6]       128     119
                     [ 7]       167     145
                     [ 8]       211     215
                     [ 9]        97     114
```

Frameworkless unit tests for:

- ReadWrite
- Grayscaling
- Sobel

# Sobel performance with Ofast




62% faster for Naive

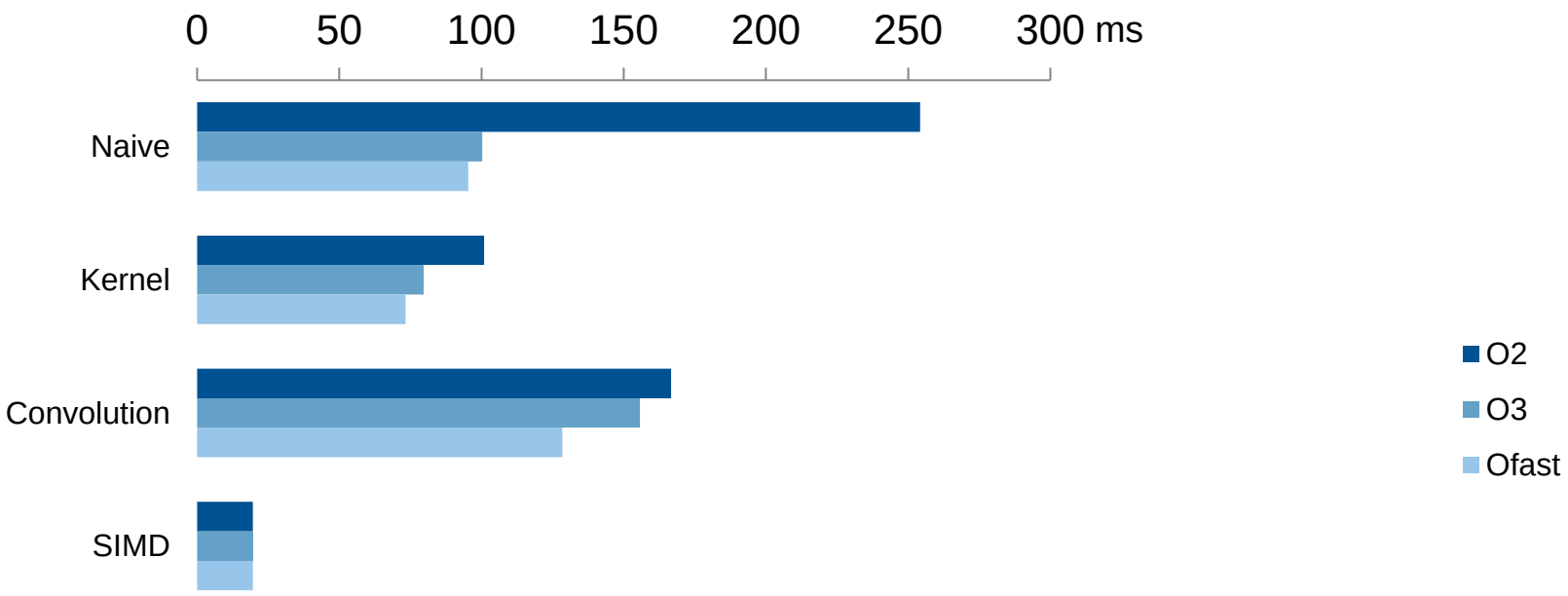
27% faster for Kernel unroll

22% faster for Convolution

no improvement for SIMD

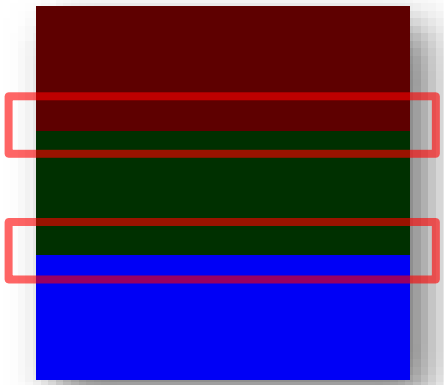
AMD Ryzen 5 4680U, 2.20 GHz, 8 GB RAM, compiled with -Ofast

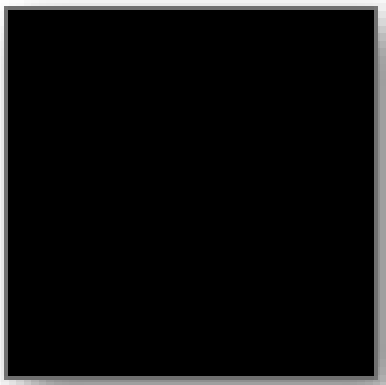
# Compiler optimization



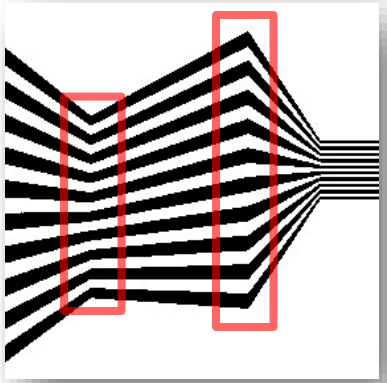

AMD Ryzen 5 4680U, 2.20 GHz, 8 GB RAM, compiled with -O2

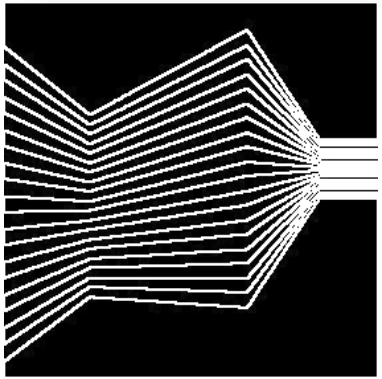
# General limitiations of the Sobel Filter





only works, if numeric representation of colour values differ





only colour based and very local, no global perception

# Code optimization

- Microoptimization not very useful, C compiler goes a long way
- SIMD reduced computation time by a lot
- Larger Kernels for more accurate results
- Threads as another possible improvement